

2 UNIT MATHEMATICS FORM VI

Time allowed: 3 hours (plus 5 minutes reading)

Exam date: 10th August, 1999

Instructions:

- All questions may be attempted.
- All questions are of equal value.
- Part marks are shown in boxes in the left margin.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection:

- Each question will be collected separately.
- Start each question in a new answer booklet.
- If you use a second booklet for a question, place it inside the first. Don't staple.
- Write your candidate number on each answer booklet.

QUESTION ONE (Start a new answer booklet)

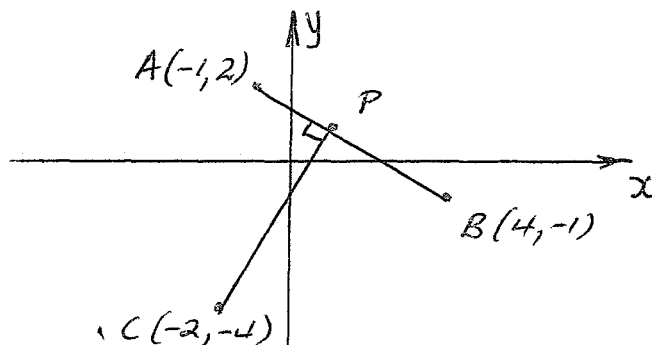
Marks

- 2** (a) Evaluate $\frac{2.3}{\sqrt[3]{2.76 - 1.09^2}}$, correct one decimal place. 1.978
- 2** (b) Find the values of x for which $|2x - 1| < 5$.
- 2** (c) Factorize completely $2x^3 - 54$.
- 2** (d) Express $\frac{3}{2\sqrt{3} - 1}$ in the form $a + b\sqrt{3}$.
- 2** (e) Simplify fully $\frac{1 - \cos^2 \theta}{\sin \theta \cos \theta}$.
- 2** (f) Find, to the nearest degree, the acute angle between the line $3x - 2y + 7 = 0$ and the x -axis.

QUESTION TWO (Start a new answer booklet)

Marks

9 (a)



In the diagram above, AB is the interval joining the points $A(-1, 2)$ and $B(4, -1)$. P is the foot of the perpendicular drawn from the point $C(-2, -4)$ to AB .

(i) Copy this diagram into your answer booklet.

(ii) Show that the distance from A to B is $\sqrt{34}$ units.

(iii) Find the gradient of the line AB and hence show that its equation is $3x + 5y - 7 = 0$.

(iv) Find the perpendicular distance from C to AB and hence find the area of $\triangle ABC$.

(v) Find the coordinates of the midpoint M of AC and show it on your diagram. Use this point to find the coordinates of point D so that $ABCD$ is a parallelogram.

3 (b) For the function $f(x) = \frac{1}{\sqrt{4 - x^2}}$ find:

(i) the domain of $f(x)$,

(ii) the range of $f(x)$.

QUESTION THREE (Start a new answer booklet)

Marks

- 5 (a) Differentiate the following with respect to x :

(i) $x^2 - \frac{1}{x^2}$,

(ii) $x^2 e^x$ (use the product rule),

(iii) $\frac{\log_e x}{x}$ (use the quotient rule).

- 3 (b) The function $y = ax^3 + bx + 4$ has a stationary point at $(1, -2)$. Write down two equations and solve them to find the values of a and b .

- 4 (c) Find:

(i) $\int \frac{1}{(x-4)^2} dx$,

(ii) $\int_5^{e+4} \frac{1}{x-4} dx$.

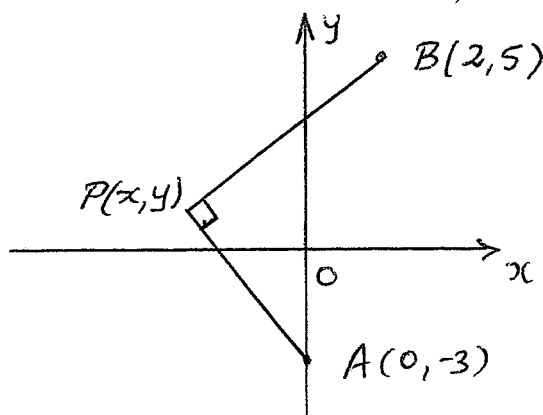
$$x^{-2}$$

$$-2x^{-3}$$

QUESTION FOUR (Start a new answer booklet)

Marks

4 (a)

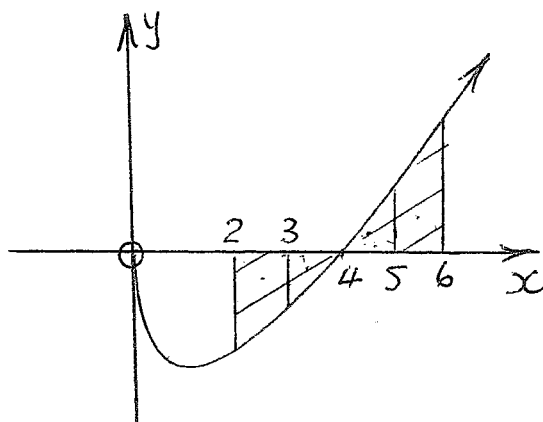


In the diagram above, points A and B have coordinates $(0, -3)$ and $(2, 5)$ respectively. $P(x, y)$ is a point such that PA is perpendicular to PB .

(i) Prove that the locus of P is the circle $x^2 + y^2 - 2x - 2y - 15 = 0$.

(ii) Find the centre and the radius of this circle.

4 (b)



The diagram above shows the graph of $y = x \log_e \left(\frac{x}{4} \right)$.

(i) Copy and complete the following table, giving your answers correct to three decimal places where necessary.

x	2	3	4	5	6
y					

(ii) By considering areas above and below the x -axis, use Simpson's rule with these five function values to evaluate the area shaded on the graph. Give your answers correct to two decimal places.

4 (c) Find the equation of the tangent to the curve $y = 1 + \cos 2x$ at the point $(\frac{\pi}{4}, 1)$. Give your answer in general form.

QUESTION FIVE (Start a new answer booklet)

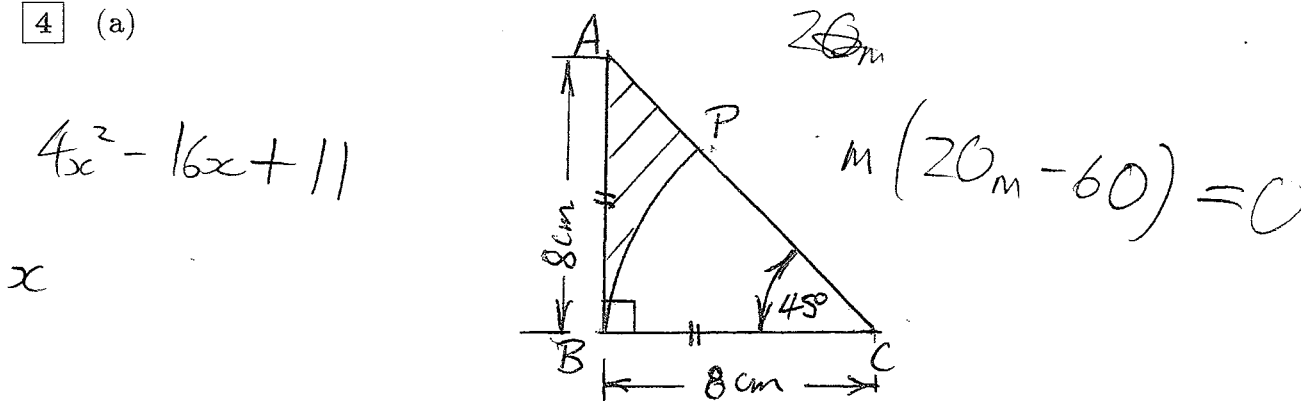
Marks

- 6** (a) The n th term of an arithmetic series is given by $T_n = 9 - 2n$.
- (i) List the first five terms and hence find the first term and the common difference.
 - (ii) Show that the sum S_n of the first n terms is $8n - n^2$.
 - (iii) Hence find the least number of terms of the series which need to be taken for this sum to be less than -945 .
- 6** (b) A quadratic function has equation $f(x) = mx^2 - 4mx - m + 15$, where m is a constant. Find the values of m for which:
- (i) ~~3~~ is a zero of $f(x)$,
 - (ii) $f(x)$ is positive definite,
 - (iii) $\alpha + \beta = \alpha\beta$, where α and β are the zeroes of $f(x)$.

QUESTION SIX (Start a new answer booklet)

Marks

- 4** (a)



In the diagram above, ABC is a right-angled isosceles triangle with $\angle ABC = 90^\circ$ and $AB = BC = 8$ cm. Arc BP with centre C and radius CB is drawn to meet AC in P .

Find, in exact form:

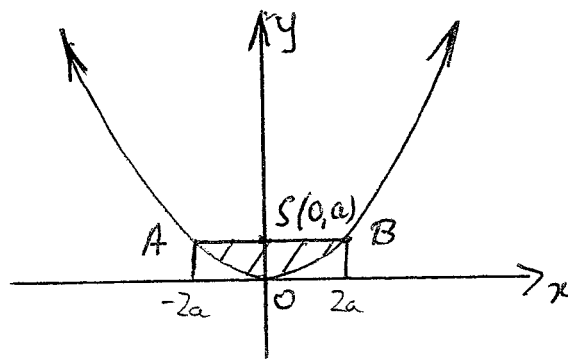
- (i) the area of the shaded region ABP ,
- (ii) the perimeter of the shaded region.

- 8 (b) The function $y = f(x)$ is given by the equation $y = \frac{1}{3}x^3 - x^2 + 1$.
- Find any stationary points and determine their nature.
 - Find any points of inflexion.
 - Sketch $y = f(x)$ in the domain $-2 \leq x \leq 3$ giving coordinates of all turning points, points of inflexion and end-points. You need not find the x -intercepts.

QUESTION SEVEN (Start a new answer booklet)

Marks

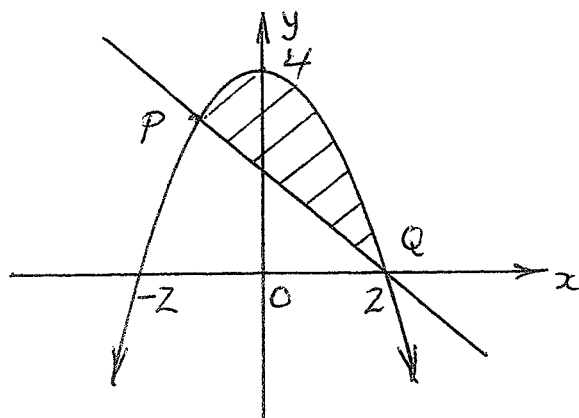
- 4 (a)



The diagram shows the graph of the parabola $x^2 = 4ay$. The interval AB is the focal chord that is parallel to the directrix and has equation $y = a$.

- Find the coordinates of the points A and B .
- Find the area enclosed by the focal chord AB and the parabola.

- 4 (b)



In the diagram above, the functions $y = 4 - x^2$ and $y = 2 - x$ intersect in the points P and Q .

- By solving these equations simultaneously, show that the x -values at P and Q are -1 and 2 respectively.
- Find the volume generated when the area enclosed by the two functions is rotated about the x -axis.

- 4 (c) The population of a small country town is growing at a rate that is proportional to the number of people in the town. The population P after t years is therefore $P = P_0 e^{kt}$, where k is a constant and P_0 is the initial population.

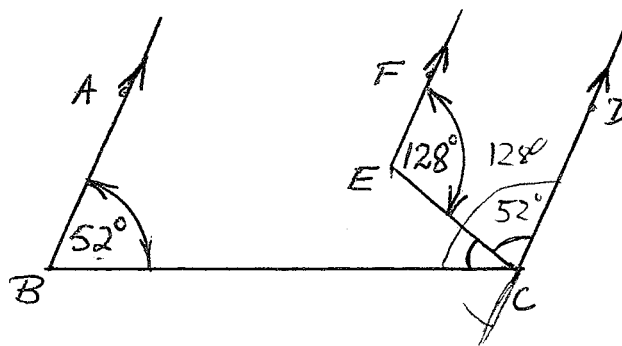
If the initial population is 6000 and ten years later the population is 9000 find:

- the value of k in exact form,
- how many years (to the nearest whole number) it will take for the population to reach five times its initial value.

QUESTION EIGHT (Start a new answer booklet)

Marks

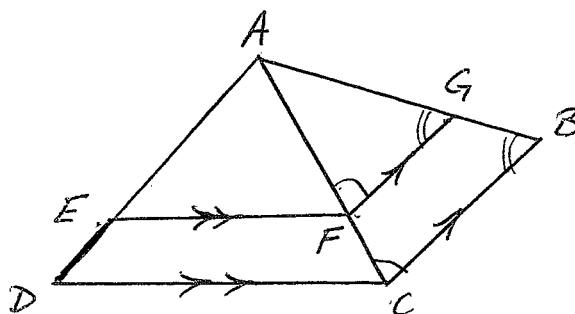
- 3 (a)



In the diagram above, AB , EF and CD are parallel lines. $\angle ABC = 52^\circ$ and $\angle FEC = 128^\circ$.

Find the size of $\angle BCE$ stating all reasons.

- 3 (b)

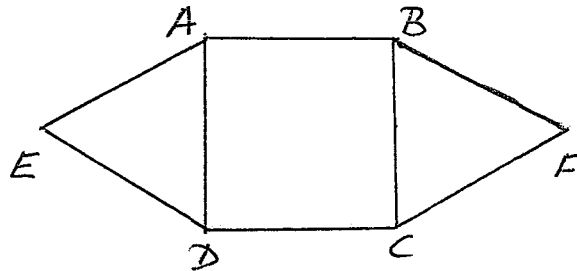


In the diagram above, EF is parallel to DC and FG is parallel to CB .

- Copy this diagram into your answer booklet.
- If $AG = 6$ cm, $AB = 9$ cm and $AE = 8$ cm, show this information on your diagram and find, stating all reasons, the length of the interval ED .

$$AG : AB = AF : AC$$

6 (c)



In the diagram above, $ABCD$ is a square, and equilateral triangles AED and BFC have been constructed on the sides AD and BC respectively.

Copy the diagram into your answer booklet and use it to prove that:

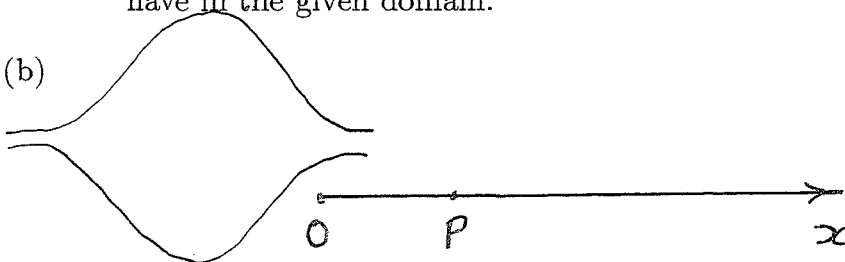
- (i) $\triangle ABF \equiv \triangle CDE$,
- (ii) $AF = EC$,
- (iii) $AFCE$ is a parallelogram.

QUESTION NINE (Start a new answer booklet)

Marks

- 4 (a) (i) Sketch on the same number plane the graphs of $y = 3 \sin 2x$ and $y = 1 - \cos x$, for $0 \leq x \leq 2\pi$.
- (ii) Hence determine the number of solutions the equation $3 \sin 2x + \cos x = 1$ will have in the given domain.

8 (b)



In the diagram above, the particle P is moving from rest from a fixed point O in the positive direction. The displacement x metres of the particle from O at time t seconds is given by:

$$x = 30t - 150 + 150e^{-0.2t}.$$

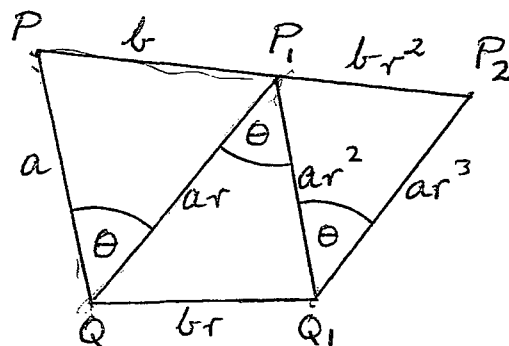
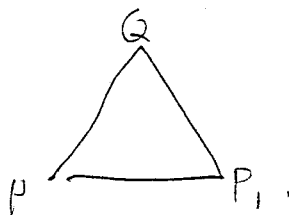
- (i) Show that its velocity at time t seconds is:

$$v = 30(1 - e^{-0.2t}).$$
- (ii) Explain why the velocity will never exceed 30 metres per second.
- (iii) Find after what time, to the nearest 0.1 second, the particle will attain a velocity of 15 metres per second, and find its displacement, to the nearest metre, at that time.
- (iv) Find the acceleration of the particle at O .

QUESTION TEN (Start a new answer booklet)

Marks

4 (a)



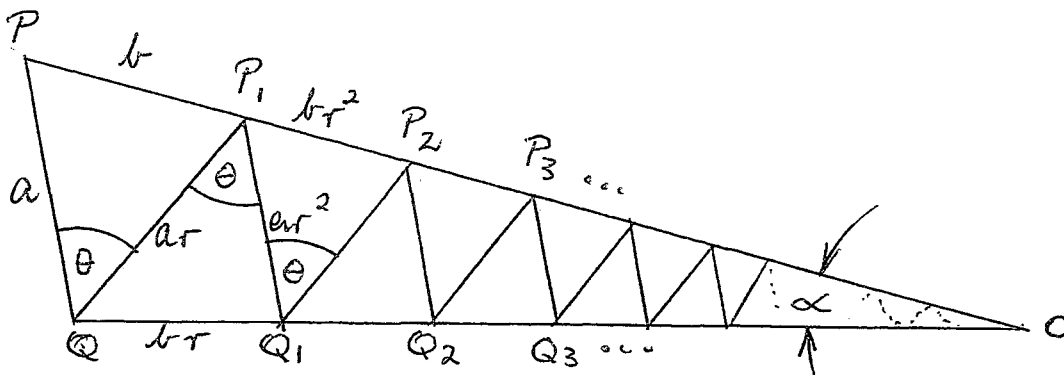
The diagram above shows three similar triangles:

$$\triangle PQP_1 \parallel \triangle QP_1Q_1 \parallel \triangle P_1Q_1P_2$$

with $\angle PQP_1 = \angle QP_1Q_1 = \angle P_1Q_1P_2 = \theta$. The lengths of the sides PQ , QP_1 , P_1Q_1 and Q_1P_2 are a , ar , ar^2 and ar^3 respectively and form a geometric sequence where $0 < r < 1$. Also, sides PP_1 , QQ_1 and P_1P_2 have length b , br and br^2 respectively.

- Copy the diagram into your answer booklet and use the fact that the triangles are similar to prove that the points P , P_1 and P_2 are collinear.
- Show that the area of $\triangle PQP_1$ is $\frac{1}{2}ra^2 \sin \theta$, and hence show that the ratio of the area of $\triangle QP_1Q_1$ to the area of $\triangle PQP_1$ is r^2 .

8 (b)



The pattern established in part (a) is continued as shown above to form an infinite sequence of similar triangles PQP_1 , QP_1Q_1 , $P_1Q_1P_2$ Let the lines PP_1P_2 ... and QQ_1Q_2 ... meet at O , and let $\angle QOP = \alpha$.

- Use the sum of an infinite sequence to find the lengths of OP and OQ .
- Also using infinite sequences, show that the area of $\triangle QOP$ is $\frac{ra^2 \sin \theta}{2(1-r^2)}$.
- Using parts (i) and (ii) above, prove that $\frac{\sin \alpha}{\sin \theta} = \frac{a^2(1-r^2)}{b^2}$.
- When $\theta = 60^\circ$ and $r = 0.9$, it can be shown by the cosine rule that $b = \frac{a\sqrt{91}}{10}$ (you need not prove this). Find the value of α to the nearest minute.

GJ

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

QUESTION ONE

(a) $\frac{2.3}{\sqrt[3]{2.76 - 1.09^2}} = 1.978\dots$
 $\div 2.0$ (to one decimal place) $\sqrt{\sqrt{}}$ (2 marks correct answer)

(b) $|2x - 1| < 5$.
The distance from x to $\frac{1}{2}$ is less than $\frac{5}{2}$.

So $-2 < x < 3$ $\sqrt{\sqrt{}}$ (-1 each error)

(c) $2x^3 - 54 = 2(x^3 - 27)$
 $= 2(x - 3)(x^2 + 3x + 9)$ $\sqrt{\sqrt{}}$ (-1 each error)

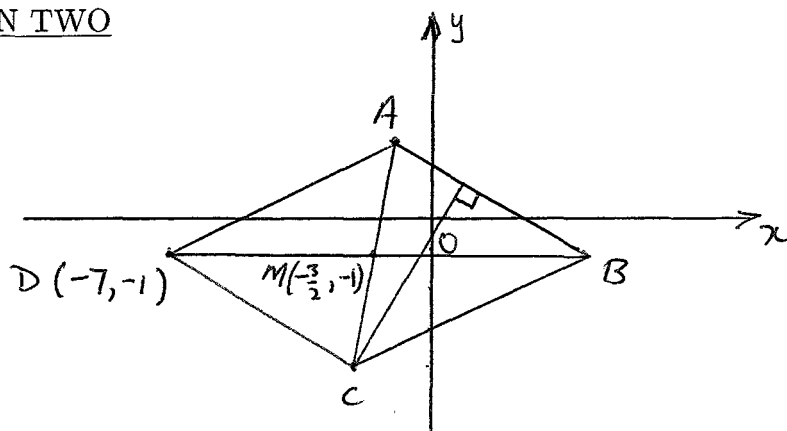
(d) $\frac{3}{2\sqrt{3} - 1} = \frac{3}{2\sqrt{3} - 1} \times \frac{2\sqrt{3} + 1}{2\sqrt{3} + 1}$ $\sqrt{}$
 $= \frac{3(2\sqrt{3} + 1)}{12 - 1}$
 $= \frac{3}{11} + \frac{6\sqrt{3}}{11}$ $\sqrt{}$

(e) $\frac{1 - \cos^2 \theta}{\sin \theta \cos \theta} = \frac{\sin^2 \theta}{\sin \theta \cos \theta}$ $\sqrt{}$
 $= \tan \theta$ $\sqrt{}$

(f) $3x - 2y + 7 = 0$
 $\tan \theta = \frac{3}{2}$ $\sqrt{}$
 $\theta \div 56^\circ$ (to nearest degree) $\sqrt{}$

QUESTION TWO

(a) (i)



(ii) $d = \sqrt{25 + 9}$
 $= \sqrt{34}$ units. ☒

(iii) gradient $= -\frac{3}{5}$ ☒
 so $y - 2 = -\frac{3}{5}(x + 1)$
 $3x + 5y - 7 = 0$. ☒

(iv) $p = \left| \frac{-6 - 20 - 7}{\sqrt{9 + 25}} \right|$ ☒
 $= \frac{33}{\sqrt{34}}$ units. ☒
 Area $= \frac{1}{2} \times \frac{33}{\sqrt{34}} \times \sqrt{34}$ ☒
 $= 33$ units². ☒

(v) Midpoint $M = (-\frac{3}{2}, -1)$. ☒

From the diagram, $D = (-7, -1)$ since the diagonals bisect each other. ☒

The reason must be given.

The alternative is to use the midpoint formula again.

(b) (i) Domain is $-2 < x < 2$ ☒

(ii) Range is $y > \frac{1}{2}$ ☒

QUESTION THREE

$$\begin{aligned}
 \text{(a) (i)} \quad y &= x^2 - \frac{1}{x^2} \\
 &= x^2 - x^{-2} \\
 \frac{dy}{dx} &= 2x + 2x^{-3} \\
 &= 2x + \frac{2}{x^3}. \quad \boxed{\sqrt{\sqrt{(-1 \text{ each error, accept negative index})}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad y &= x^2 e^x \\
 \frac{dy}{dx} &= 2x e^x + x^2 e^x. \quad \boxed{\sqrt{}}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad y &= \frac{\log_e x}{x} \\
 \frac{dy}{dx} &= \frac{x \times \frac{1}{x} - \log_e x}{x^2} \\
 &= \frac{1 - \log_e x}{x^2}. \quad \boxed{\sqrt{\sqrt{(-1 \text{ each error})}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad y &= ax^3 + bx + 4 \\
 \text{so} \quad -2 &= a + b + 4 \\
 a + b &= -6. \quad \boxed{\sqrt{}} \quad (1)
 \end{aligned}$$

$$\text{Also} \quad \frac{dy}{dx} = 3ax^2 + b$$

$$\text{and} \quad \frac{dy}{dx} = 0 \text{ at } x = 1$$

$$\text{so} \quad 3a + b = 0. \quad \boxed{\sqrt{}} \quad (2)$$

$$(2) - (1) \quad 2a = 6$$

$$\text{so} \quad a = 3$$

$$\text{and} \quad b = -9. \quad \boxed{\sqrt{}}$$

$$\begin{aligned}
 \text{(c) (i)} \quad \int \frac{dx}{(x-4)^2} &= \int (x-4)^{-2} dx \quad \boxed{\sqrt{}} \\
 &= \frac{-1}{(x-4)} + c, \text{ (where } c \text{ is a constant)} \quad \boxed{\sqrt{}}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \int_5^{e+4} \frac{dx}{x-4} &= \left[\log_e(x-4) \right]_5^{e+4} \quad \boxed{\sqrt{}} \\
 &= \log_e e - \log_e 1 \\
 &= 1 \quad \boxed{\sqrt{}}
 \end{aligned}$$

QUESTION FOUR

$$(a) \quad (i) \quad \frac{y-5}{x-2} \times \frac{y+3}{x} = -1 \quad \boxed{\checkmark}$$

$$x(x-2) + (y-5)(y+3) = 0$$

$$x^2 - 2x + y^2 - 2y - 15 = 0$$

$$x^2 + y^2 - 2x - 2y - 15 = 0 \quad \boxed{\checkmark}$$

$$(ii) \quad x^2 - 2x + 1 + y^2 - 2y + 1 = 17$$

$$(x-1)^2 + (y-1)^2 = 17.$$

The centre is (1, 1) and the radius is $\sqrt{17}$ units. $\boxed{\checkmark\checkmark}$

$$(b) \quad (i) \quad \begin{array}{|c|c|c|c|c|c|} \hline x & 2 & 3 & 4 & 5 & 6 \\ \hline y & -1.386 & -0.863 & 0 & 1.116 & 2.433 \\ \hline \end{array} \quad \boxed{\checkmark}$$

$$\text{Area} \div \left| \frac{4-2}{6}(-1.386 + 4 \times -0.863) \right| + \frac{6-4}{6}(4 \times 1.116 + 2.433) \quad \boxed{\checkmark\checkmark}$$

$$\div 3.91 \text{ square units (to 2 decimal places)} \quad \boxed{\checkmark}$$

$$(c) \quad y = 1 + \cos 2x$$

$$\frac{dy}{dx} = -2 \sin 2x \quad \boxed{\checkmark}$$

$$\text{gradient} = -2 \sin \frac{\pi}{2} \quad \boxed{\checkmark}$$

$$y - 1 = -2\left(x - \frac{\pi}{4}\right) \quad \boxed{\checkmark}$$

$$y - 1 = -2x + \frac{\pi}{2}$$

$$4x + 2y - 2 - \pi = 0 \quad \boxed{\checkmark}$$

QUESTION FIVE

$$(a) \quad (i) \quad T_n = 9 - 2n$$

$$T_1 = 7, T_2 = 5, T_3 = 3, T_4 = 1, T_5 = -1,$$

$$\text{so } a = 7 \quad \boxed{\checkmark}$$

$$\text{and } d = -2. \quad \boxed{\checkmark}$$

$$(ii) \quad S_n = \frac{n}{2}[2a + (n-1)d]$$

$$= \frac{n}{2}[14 + (n-1) \times -2]$$

$$= \frac{n}{2}(16 - 2n)$$

$$= n(8 - n)$$

$$= 8n - n^2. \quad \boxed{\checkmark}$$

(iii) Put $S_n < -945$

$$8n - n^2 < -945 \quad \boxed{\checkmark}$$

$$n^2 - 8n - 945 > 0$$

$$(n - 35)(n + 27) > 0$$

so $n < -27$ or $n > 35$. $\boxed{\checkmark}$

Since $n > 0$ the least number of terms required is 36. $\boxed{\checkmark}$

(b) (i) $f(x) = mx^2 - 4mx - m + 15$

$$f(3) = 0$$

$$0 = 9m - 12m - m + 15$$

$$4m = 15$$

$$m = \frac{15}{4}. \quad \boxed{\checkmark}$$

(ii) We require $m > 0$ and $\Delta < 0$, $\boxed{\checkmark}$

$$\Delta = 16m^2 - 4m(-m + 15)$$

$$= 16m^2 + 4m^2 - 60m$$

$$= 20m^2 - 60m$$

$$= 20m(m - 3).$$

Now $20m(m - 3) < 0 \quad \boxed{\checkmark}$

so $0 < m < 3$. $\boxed{\checkmark}$

(iii) Put $-\frac{b}{a} = \frac{c}{a} \quad \boxed{\checkmark}$

so $-b = c$

$$4m = -m + 15$$

$$5m = 15$$

$$m = 3. \quad \boxed{\checkmark}$$

QUESTION SIX

(a) (i) Area $\triangle ABC = \frac{1}{2} \times 8 \times 8$
 $= 32 \text{ cm}^2.$

$$\text{Area sector} = \frac{1}{2} \times 64 \times \frac{\pi}{4}$$

$$= \frac{32\pi}{4} \text{ cm}^2.$$

$$\text{Shaded area} = 32 - \frac{32\pi}{4}$$

$$= 8(4 - \pi) \text{ cm}^2. \quad \boxed{\sqrt{\checkmark} (-1 \text{ each error})}$$

(ii) Length $AC = 8\sqrt{2}$ cm.

Length $AP = 8\sqrt{2} - 8$ cm.

Length arc $PB = 8 \times \frac{\pi}{4}$

$= 2\pi$ cm.

Perimeter $= 8\sqrt{2} - 8 + 8 + 2\pi$

$= 8\sqrt{2} + 2\pi$ cm. $\boxed{\sqrt{\sqrt{(-1 \text{ each error})}}}$

(b) (i) $y = \frac{1}{3}x^3 - x^2 + 1$

$\frac{dy}{dx} = x^2 - 2x$

Put $x^2 - 2x = 0$ for stationary points

$x(x - 2) = 0$

so $x = 0$ or 2 .

The stationary points are $(0, 1)$ and $(2, -\frac{1}{3})$. $\boxed{\checkmark}$

When $x = 0$ $\frac{d^2y}{dx^2} = -2$, so $(0, 1)$ is a maximum turning point. $\boxed{\checkmark}$

When $x = 2$ $\frac{d^2y}{dx^2} = 2$, so $(2, -\frac{1}{3})$ is a minimum turning point. $\boxed{\checkmark}$

(ii) $\frac{d^2y}{dx^2} = 0$ for points of inflexion

so $2x - 2 = 0$

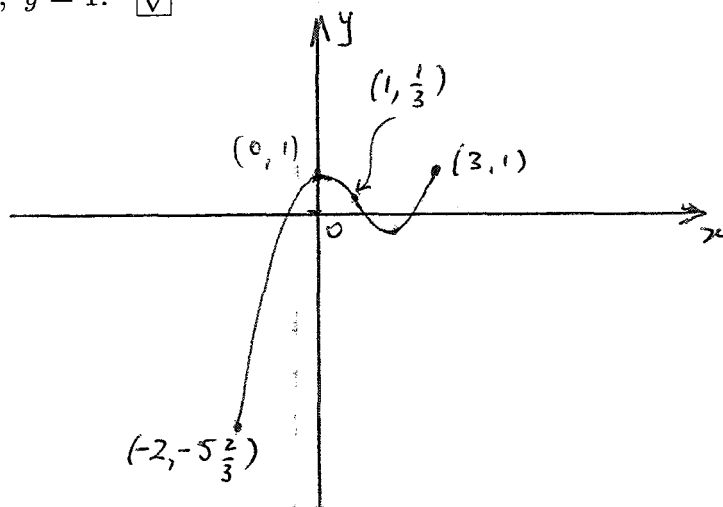
$x = 1$. $\boxed{\checkmark}$

x	0	1	2
$\frac{d^2y}{dx^2}$	-2	0	2
	$\cap$	.	$\cup$

So the point of inflexion is verified at $(1, \frac{1}{3})$. $\boxed{\checkmark}$

(iii) When $x = -2$, $y = -5\frac{2}{3}$,

and when $x = 2$, $y = 1$. $\boxed{\checkmark}$



$\boxed{\checkmark\checkmark}$

QUESTION SEVEN

$$(a) \quad (i) \quad x^2 = 4ay \quad (1)$$

$$y = a \quad (2)$$

$$\text{so } x^2 = 4a^2$$

$$x = -2a \text{ or } 2a.$$

$$\text{So } A \text{ is } (-2a, a) \text{ and } B \text{ is } (2a, a). \quad \boxed{\checkmark}$$

$$(ii) \quad y = \frac{x^2}{4a},$$

$$\text{so area} = \int_{-2a}^{2a} \left(a - \frac{x^2}{4a}\right) dx \quad \boxed{\checkmark}$$

$$= 2 \int_0^{2a} \left(a - \frac{x^2}{4a}\right) dx$$

$$= 2 \left[ax - \frac{x^3}{12a} \right]_0^{2a} \quad \boxed{\checkmark}$$

$$= 2 \left(2a^2 - \frac{8a^2}{12a} \right)$$

$$= \frac{8a^2}{3} \text{ units}^2. \quad \boxed{\checkmark}$$

$$(b) \quad (i) \quad y = 4 - x^2 \quad (1)$$

$$y = 2 - x \quad (2)$$

$$(1) - (2) \quad 0 = 2 - x^2 + x$$

$$x^2 - x - 2 = 0$$

$$(x - 2)(x + 1) = 0$$

$$\text{so } x = -1 \text{ or } 2.$$

$$\text{At } P, x = -1 \text{ and at } Q, x = 2. \quad \boxed{\checkmark}$$

$$(ii) \quad V = \pi \int_{-1}^2 (4 - x^2)^2 + (2 - x)^2 dx \quad \boxed{\checkmark}$$

$$= \pi \int_{-1}^2 (x^4 - 9x^2 + 4x + 12) dx$$

$$= \pi \left[\frac{x^5}{5} - 3x^3 + 2x^2 + 12x \right]_{-1}^2 \quad \boxed{\checkmark}$$

$$= \pi \left[\frac{32}{5} - 24 + 8 + 24 \right] - \pi \left[-\frac{1}{5} + 3 + 2 - 12 \right]$$

$$= \frac{108\pi}{5} \text{ units}^3. \quad \boxed{\checkmark}$$

$$(c) \quad (i) \quad P = P_0 e^{kt}$$

$$9000 = 6000e^{10k}$$

$$k = \frac{1}{10} \log_e \frac{3}{2}. \quad \boxed{\checkmark}$$

$$(ii) 30000 = 6000e^{t\left(\frac{1}{10}\log_e \frac{3}{2}\right)} \quad \boxed{\checkmark}$$

$$\log_e 5 = t\left(\frac{1}{10}\log_e \frac{3}{2}\right) \quad \boxed{\checkmark}$$

$$t = \frac{10\log_e 5}{\log_e \frac{3}{2}}$$

$$\div 37 \text{ years, (to the nearest whole number).} \quad \boxed{\checkmark}$$

QUESTION EIGHT

$$(a) (i) \angle DCE = 52^\circ \text{ (cointerior angles, } FE \parallel DC) \quad \boxed{\checkmark}$$

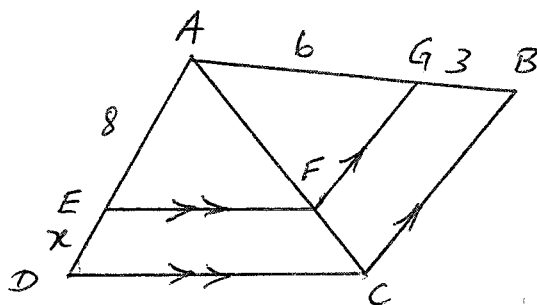
$$\angle DCE = 128^\circ \text{ (cointerior angles, } AB \parallel DC) \quad \boxed{\checkmark}$$

$$\angle BCE = \angle BCD - \angle DCE$$

$$= 128^\circ - 52^\circ$$

$$= 76^\circ. \quad \boxed{\checkmark}$$

(b) (i)



$$(ii) \frac{AG}{GB} = \frac{AF}{FC}$$

$$\frac{AF}{FC} = \frac{AE}{ED} \text{ (intercept properties on transversals)} \quad \boxed{\checkmark}$$

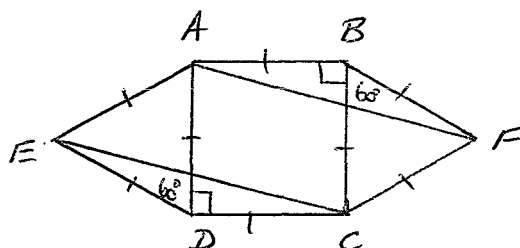
$$\text{so } \frac{AG}{GB} = \frac{AE}{ED} \quad \boxed{\checkmark}$$

$$\frac{6}{3} = \frac{8}{x}$$

$$x = 4.$$

$$\text{So } ED = 4 \text{ cm.} \quad \boxed{\checkmark}$$

(c) (i)



In square $ABCD$:

$AB = BC = CD = DA$ (equal sides). ☒

In equilateral triangles BFC and ADE :

$BC = CF = FB = AD = AE = ED$ (equal sides of equilateral triangles and $AD = BC$).

So $BF = ED$ and $AB = CD$. ☒

Also $\angle ABF = \angle ABC + \angle CBF$

$$= 90^\circ + 60^\circ \text{ (sum interior angle of a square and an equilateral triangle)}$$

$$= 150^\circ$$

Similarly $\angle EDC = 150^\circ$. ☒

Join EC and AF .

In $\triangle ABF$ and $\triangle EDC$,

1. $BF = ED$ from above

2. $AB = CD$ from above

3. $\angle ABF = \angle EDC$ from above

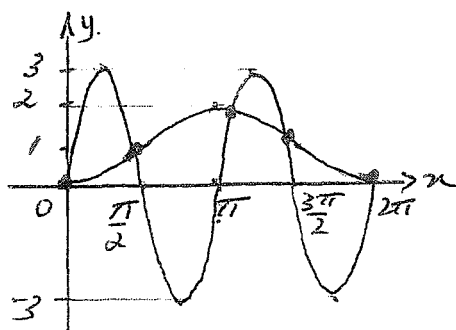
so $\triangle ABF \equiv \triangle EDC$ (SAS). ☒

(ii) $AF = EC$ (matching sides of congruent triangles). ☒

(iii) Now $AF = EC$ and $AE = FC$ so $AFCE$ is a parallelogram (opposite sides equal). ☒

QUESTION NINE

(a) (i)



☒ for sine curve

☒ for cosine curve

(ii) There are five solutions. ☒

(b) (i) $x = 30t - 150 + 150e^{-0.2t}$

$v = 30 - 30e^{-0.2t}$

$v = 30(1 - e^{-0.2t})$ ☒

(ii) As $t \rightarrow \infty$, $e^{-0.2t} \rightarrow 0$ so $v \rightarrow 30$ from below.

The velocity does not exceed 30 metres per second. ☒

(iii) Let $v = 15$

so $15 = 30(1 - e^{-0.2t})$ ☒

$e^{-0.2t} = \frac{1}{2}$

$t = -\frac{1}{0.2} \log_e \frac{1}{2}$

$\doteq 3.5$ seconds (to the nearest 0.1 second). ☒

$x = 30 \times 3.5 - 150 + 150e^{-0.2 \times 3.5}$

$\doteq 29$ metres (to the nearest metre). ☒

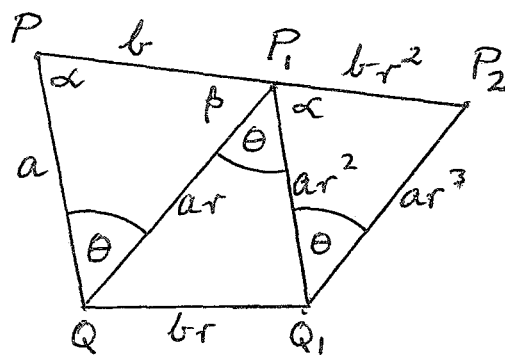
(iv) $\frac{d^2y}{dx^2} = 6e^{-0.2t}$ ☒

$= 6$ when $t = 0$.

So the initial acceleration is 6 m/sec². ☒

QUESTION TEN

(a) (i)



Let $\angle QPP_1 = \alpha$ and $\angle QP_1P = \beta$

so $\theta + \alpha + \beta = 180^\circ$ (angle sum of triangle). ☒

Now $\triangle PP_1Q \parallel \triangle P_1P_2Q_1$

so $\angle QPP_1 = \angle Q_1P_1P_2 = \alpha$,

so $\angle PP_1Q + \angle QP_1Q_1 + \angle Q_1P_1P_2 = \alpha + \theta + \beta$
 $= 180^\circ$.

So P, P₁ and P₂ are collinear. ☒

$$\begin{aligned}
 \text{(ii)} \quad \text{Area } \triangle PQP_1 &= \frac{1}{2} \times a \times ar \times \sin \theta \\
 &= \frac{1}{2} a^2 r \sin \theta. \quad \boxed{\checkmark} \\
 \text{Area } \triangle QP_1Q_1 &= \frac{1}{2} \times ar \times ar^2 \times \sin \theta \\
 &= \frac{1}{2} a^2 r^3 \sin \theta \\
 \text{so } \frac{\text{Area } \triangle QP_1Q_1}{\text{Area } \triangle PQP_1} &= \frac{\frac{1}{2} a^2 r^3 \sin \theta}{\frac{1}{2} a^2 r \sin \theta} \\
 &= r^2. \quad \boxed{\checkmark}
 \end{aligned}$$

(b) (i) Length of OP is the limiting sum of the lengths $PP_1, P_1P_2, P_2P_3, \dots$

First term = b ,

common ratio = r^2

$$\text{so} \quad OP = \frac{b}{1 - r^2}. \quad \boxed{\checkmark}$$

$$\text{Similarly,} \quad OQ = \frac{br}{1 - r^2}. \quad \boxed{\checkmark}$$

(ii) Area of $\triangle QOP$ is the limiting sum of the areas of the triangles $PQP_1, QP_1Q_1, P_1Q_1P_2, \dots$

First term = $\frac{1}{2} r a^2 \sin \theta$,

common ratio = r^2 .

$$\begin{aligned}
 \text{Area } \triangle QOP &= \frac{\frac{1}{2} r a^2 \sin \theta}{1 - r^2} \\
 &= \frac{r a^2 \sin \theta}{2(1 - r^2)} \quad \boxed{\checkmark}
 \end{aligned}$$

(iii) Also, area $\triangle QOP = \frac{1}{2} \times \frac{b}{1 - r^2} \times \frac{br}{1 - r^2} \times \sin \alpha$

$$= \frac{b^2 r \sin \alpha}{2(1 - r^2)^2}. \quad \boxed{\checkmark}$$

$$\text{so} \quad \frac{b^2 r \sin \alpha}{2(1 - r^2)^2} = \frac{r a^2 \sin \theta}{2(1 - r^2)} \quad \boxed{\checkmark}$$

$$\text{so} \quad \frac{\sin \alpha}{\sin \theta} = \frac{a^2(1 - r^2)}{b^2} \quad \boxed{\checkmark}$$

$$(iv) \quad b = \frac{a\sqrt{91}}{10}$$

$$\text{so} \quad b^2 = \frac{91a^2}{100}.$$

$$\text{Now } \sin \alpha = \frac{a^2(1 - r^2) \sin \alpha}{b^2}$$

$$\text{so} \quad \sin \alpha = \frac{100(1 - 0.81) \sin 60^\circ}{91}$$

$$= \frac{19\sqrt{3}}{182}. \quad \boxed{\checkmark}$$

$$\text{So} \quad \alpha \doteq 10^\circ 25' \text{ (to the nearest minute)} \quad \boxed{\checkmark}$$

GJ